

Day 7 — Why Eclipses Happen

Module: Moon Phases and Tithi · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: explain solar/lunar eclipses using orbital plane geometry (5° tilt of moon's orbit).

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *pūrṇimā* (IAST: pūrṇimā; lit. “full moon”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **Geometric setup.** Solar eclipse — Moon between Sun and Earth (occurs at *amāvasyā*). Lunar eclipse — Earth between Sun and Moon (occurs at *pūrṇimā*).
2. **Why not every new/full moon?** The Moon’s orbital plane is tilted ~5.14° from the ecliptic. Eclipses happen only when the Moon is near a *node* (intersection of orbital planes) during new or full moon. This is the *Rāhu/Ketu* node geometry of classical Indian astronomy.
3. **Source.** Āryabhaṭīya, *Gola-pāda* — gives a heliocentric-leaning model and correctly predicts eclipses 1000+ years before modern astronomy.
4. **Show a recent eclipse path map** (any annular or total eclipse from the last 5 years). “*Why does the path strike only a thin ribbon, not the whole hemisphere?*” (Geometric umbra/penumbra.)
5. **Safety rule.** Solar eclipse — NEVER look at the Sun directly. Use projection or eclipse-rated filters only. Lunar eclipse is always safe. State this rule on the board and have students repeat.

Activity (15 min)

A scaled version of *10-Day Moon Observation Diary* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Why Eclipses Happen.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).